

Two elementary analytic lemmas for stable polarization

Independent derivations supporting the power lower-bound proof

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These are analytic lemmas, not by themselves a bound on dice sizes.

Lemma 1 (Centered coefficients from variance alone). *Let $y_1, \dots, y_N$ be real, $\sum_i y_i = 0$, and $V = \sum_i y_i^2$. For $1 \leq k \leq N$,*

$$|e_k(y)| \leq (eV/k)^{k/2}, \quad \frac{|e_k(y)|}{\binom{N}{k}} \leq \left(\frac{\sqrt{ekV}}{N} \right)^k.$$

No upper bound on the largest coordinate, and no restriction $k \leq N/2$, is required.

Proof. For every complex w ,

$$|1 + w|^2 = 1 + 2 \operatorname{Re} w + |w|^2 \leq \exp(2 \operatorname{Re} w + |w|^2).$$

Consequently, for every complex z ,

$$\left| \prod_i (1 + zy_i) \right| \leq \exp(V|z|^2/2).$$

If $V > 0$, Cauchy's coefficient bound on the circle of radius $\sqrt{k/V}$ gives the first inequality. The case $V = 0$ is trivial. The second follows from $\binom{N}{k} \geq (N/k)^k$. $\square$

Lemma 2 (Off-diagonal Gaussian bound). *For ordinary Laguerre polynomials L_j , $0 \leq h \leq j$, and $x \geq 0$,*

$$x^{h/2} \left| \frac{d^h}{dx^h} L_j(x) \right| \leq \sqrt{\frac{j!}{(j-h)!}} e^{x/2}.$$

In particular, for $\theta > 0$, $z \geq 0$, and any finite sum $Q(z) = \sum_{j=0}^d c_j L_j(\theta z)$ with $B = \sum_j |c_j|$,

$$z^{h/2} |Q^{(h)}(z)| \leq B(\theta d)^{h/2} e^{\theta z/2}.$$

For two such sums Q_1, Q_2 , both of degree at most d ,

$$|(Q_1 Q_2)^{(k)}(z)| \leq B_1 B_2 e^{\theta z} \left(2\sqrt{\theta d/z} \right)^k \quad (z > 0).$$

Proof. Use the Hilbert space of entire functions with Gaussian norm

$$\|f\|^2 = \frac{1}{\pi} \int_{\mathbb{C}} |f(w)|^2 e^{-|w|^2} dA(w).$$

The monomials $f_j(w) = w^j / \sqrt{j!}$ are orthonormal. For real a , the operator

$$(U_a f)(w) = e^{-a^2/2 + aw} f(w - a)$$

is unitary, since $|e^{-a^2/2+aw}|^2 e^{-|w|^2} = e^{-|w-a|^2}$. Expansion of $e^{aw}(w-a)^{j-h}$, followed by extraction of its w^j coefficient, gives

$$\begin{aligned}\langle f_j, U_a f_{j-h} \rangle &= e^{-a^2/2} \sqrt{\frac{j!}{(j-h)!}} \sum_{t=0}^{j-h} \binom{j-h}{t} \frac{(-1)^t a^{h+2t}}{(h+t)!} \\ &= e^{-a^2/2} \sqrt{\frac{(j-h)!}{j!}} a^h L_{j-h}^{[h]}(a^2).\end{aligned}$$

Here square brackets denote the associated-Laguerre parameter, to distinguish it from differentiation. The coefficient identity is the defining expansion $L_{j-h}^{[h]}(x) = \sum_t (-1)^t \binom{j}{h+t} x^t / t!$. The matrix coefficient has absolute value at most one. Taking $a = \sqrt{x}$ and using $D^h L_j = (-1)^h L_{j-h}^{[h]}$ proves the first bound. The scaled version follows by differentiation, summation, and $j!/(j-h)! \leq d^h$. Derivatives with $h > d$ vanish. Finally the product rule and the binomial sum $\sum_h \binom{k}{h} = 2^k$ give the last assertion. At zero, only the weighted or limiting forms of these statements are intended; no division by zero is used. $\square$